Modeling and Numerical Solution of Damped Simple Pendulums and Educational Visualization with Manim

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Introduction

The simple pendulum is one of the most familiar systems in classical mechanics, but its apparently elementary behavior hides a relevant nonlinear problem when the angular amplitude is no longer small, i.e., when $\theta > 15^\circ$. In this regime, the approximation $\sin\theta \approx \theta$ is not sufficient to describe the motion, and dissipative effects modify both the amplitude and the energy of the system over time.

This work investigates damped simple pendulums at large amplitudes by combining experimental data, numerical integration, and scientific visualization. The analysis uses Python routines to solve the nonlinear equation of motion, extract peaks from measured signals, compare damping regimes, and produce visual simulations with Manim. The central idea is not only to analyze the motion, but also to build a bridge between the mathematical model, the measured dynamics, and an educational visual representation of phase-space evolution.

Research Question and Objectives

The project adresses the topic of how a damped pendulum behaves outside the linear small-angle regime and how this behavior can be clearly presented for teaching and analysis.

- Model the nonlinear damped pendulum without replacing $\sin\theta$ by θ.
- Estimate oscillation peaks using FFT-assisted peak detection.
- Analyze amplitude, speed, phase space, and energy decrease.
- Compare experimental trends with synthetic/numerical datasets.
- Generate educational animations and interactive visualizations with Manim and Python.

Theoretical Model

For small angular displacements, the damped pendulum can be approximated by the linear differential equation

$$\ddot{\theta} + 2\gamma\dot{\theta} + \omega_0^2\theta = 0,$$

where $\theta(t)$ is the angular displacement, γ is the damping parameter, and ω_0 is the natural angular frequency of the undamped system.

The behavior of the system depends on the relation between γ and ω_0. This leads to three characteristic damping regimes.

Underdamped regime: for $0 < \gamma < \omega_0$, the system oscillates with decreasing amplitude:

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2},$$

$$\theta(t) = e^{-\gamma t}\left[C_1\cos(\omega_d t) + C_2\sin(\omega_d t)\right].$$

Equivalently,

$$\theta(t) = Ae^{-\gamma t}\cos(\omega_d t + \phi).$$

Critically damped regime: for $\gamma = \omega_0$, the system returns to equilibrium without oscillating and in the shortest possible time:

$$\theta(t) = (C_1 + C_2 t)e^{-\gamma t}.$$

For the initial conditions

$$\theta(0) = \theta_0, \qquad \dot{\theta}(0) = 0,$$

the solution becomes

$$\theta(t) = \theta_0(1 + \gamma t)e^{-\gamma t}.$$

Overdamped regime: for $\gamma > \omega_0$, the system also returns to equilibrium without oscillating, but more slowly than in the critical case:

$$\alpha = \sqrt{\gamma^2 - \omega_0^2},$$

$$\theta(t) = C_1 e^{-(\gamma+\alpha)t} + C_2 e^{-(\gamma-\alpha)t}.$$

Using $\theta(0) = \theta_0$ and $\dot{\theta}(0) = 0$, one obtains

$$C_1 = \frac{\gamma - \alpha}{2\alpha}\theta_0, \qquad C_2 = \frac{\gamma + \alpha}{2\alpha}\theta_0.$$

For large amplitudes, the linear approximation $\sin\theta \approx \theta$ is no longer valid. Therefore, the complete nonlinear model is written as

$$\ddot{\theta} + 2\gamma\dot{\theta} + \omega_0^2\sin\theta = 0.$$

This equation is solved numerically and used to generate the time-series curves and phase-space visualizations presented in this work.

Numerical and Data-Analysis Pipeline

The nonlinear second-order equation was rewritten as a first-order system,

$$\dot{\theta} = \Omega, \qquad \dot{\Omega} = -\frac{b}{mL^2}\Omega - \frac{g}{L}\sin\theta.$$

The numerical solution was obtained in Python and compared with experimental time series processed from Tracker and PASCO Capstone®. The FFT calibrated the peak detection by estimating the dominant period $T = 1/f_0$. This period defines the minimum distance between peaks.

Experimental Setup

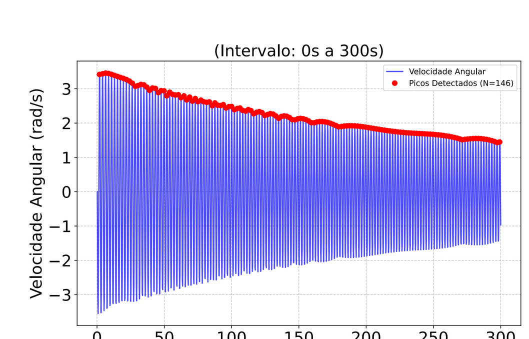
The experimental setup was designed to study the motion of a simple pendulum at large angular amplitudes. A PASCO sensor was attached to a fixed support in order to record the angular motion of the system. The pendulum consisted of a rigid rod with an approximate length of 96 cm, with a small mass attached to its lower end. This configuration allowed the pendulum to be released from large initial amplitudes, making it possible to observe nonlinear effects beyond the small-angle approximation.

(a) PASCO sensor attached to a fixed support.

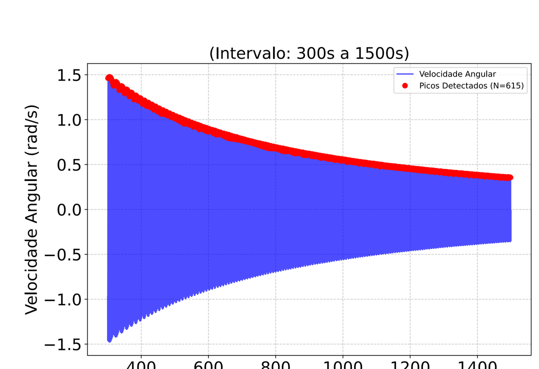
(b) Small mass attached to the end of the pendulum rod.

Speed as a Function of Time

The speed signal was analyzed in two different time intervals in order to compare the initial transient behavior with the long-time damped regime.

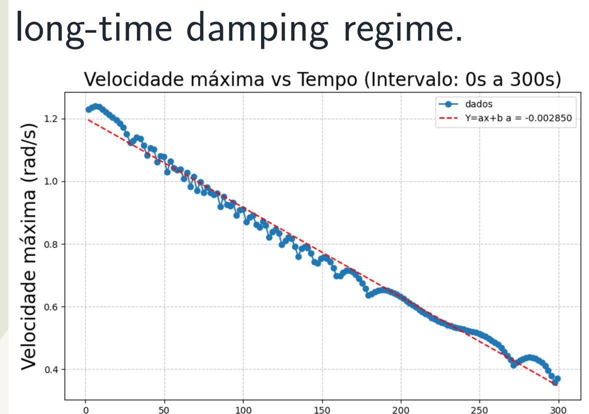


(a) 0–300 s.

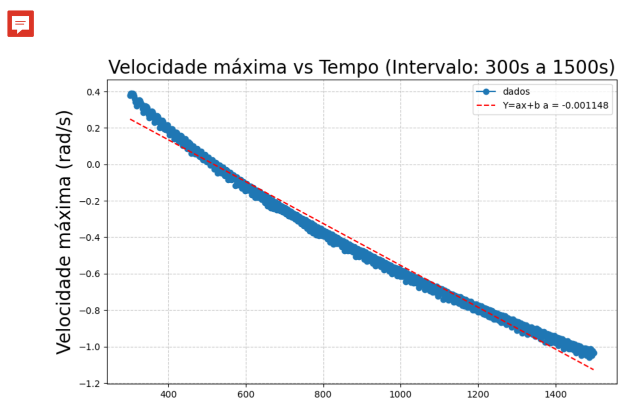


(b) 300–1500 s.

The maximum speed was extracted over time to compare the two damping regimes: a large gamma initial damping regime and a smaller gamma long-time damping regime.



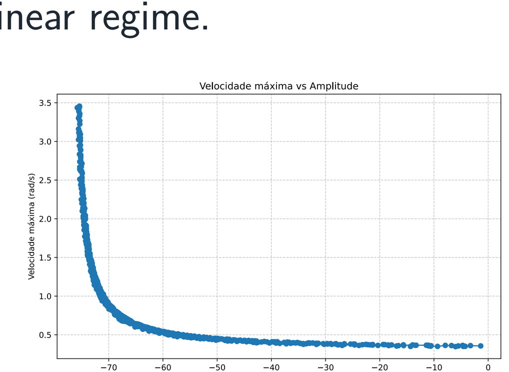
(a) Large gamma damping regime.



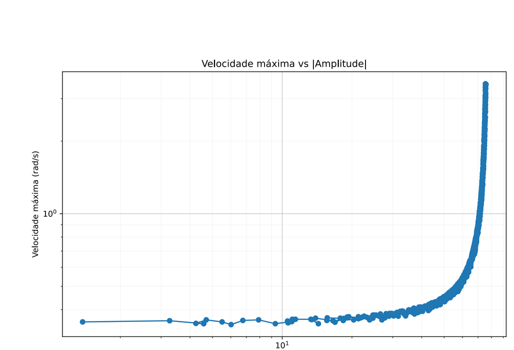
(b) Small gamma damping regime.

Speed as a Function of Amplitude

The relation between maximum speed and amplitude is used to highlight the transition from the nearly linear regime to the nonlinear large-amplitude regime. For small amplitudes, the maximum speed changes slowly, indicating that the system remains close to the linear approximation. As the amplitude increases, the maximum speed grows more rapidly, showing that the pendulum dynamics clearly depart from the linear regime.



(a) Full dataset – 0-1500s.



(b) Log–log representation.

Figure: Maximum Speed as a function of amplitude.

Damping Regimes and Phase Space

Synthetic datasets were produced for different damping values in order to study qualitative features observed in the motion: large amplitudes, energy loss, and the change of phase-space trajectories. For the underdamped regime, the trajectory forms a spiral-like curve approaching equilibrium; stronger damping suppresses oscillations and changes the geometry of the phase space trajectory.

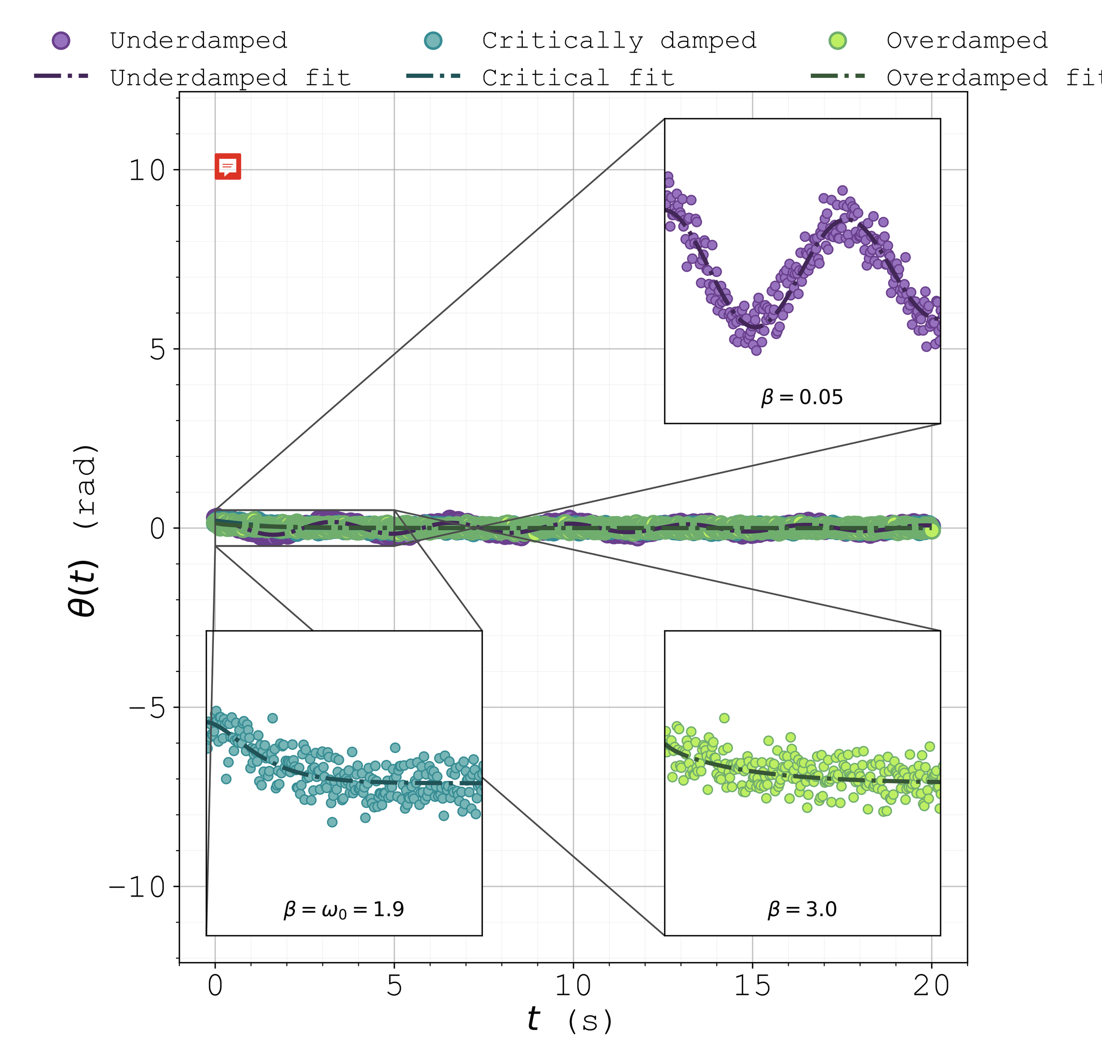


Figure: Graph showing the different damping regimes obtained from synthetic data with 5% pseudo-random Gaussian noise (seed = 0) and initial amplitude $A_0 = 0.2$ rad.

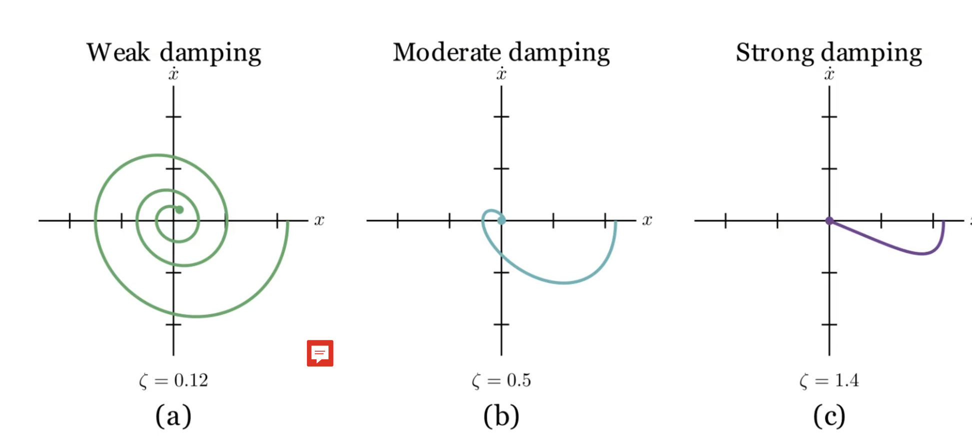


Figure: Phase-space trajectories for different damping regimes. In the weak damping regime ($\gamma = 0.12$), the system exhibits oscillatory motion with a slowly decreasing amplitude, producing a spiral trajectory towards equilibrium. For moderate damping ($\gamma = 0.5$), the oscillations decay more rapidly and the trajectory approaches equilibrium with fewer cycles. In the strong damping regime ($\gamma = 1.4$), the system returns to equilibrium without oscillating, characterizing an overdamped response.

Phase Flow and Educational Visualization

Por fazer...

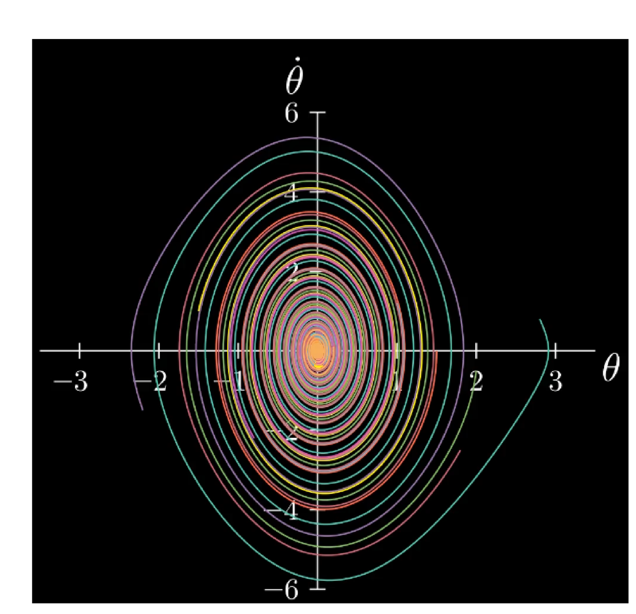


Figure: Phase-flow simulation with a set of 10 different initial conditions.

Conclusions

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Acknowledgments

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